

Little Red Dots: Super-Eddington Wind-Shrouded Engines

Type IIIn supernovae and eta Carinae's Great Eruption offer a unified model for JWST's enigmatic Little Red Dots.

arXiv:2606.30711 — Little Red Dots as intermediate-mass, super-Eddington engines.

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Background

JWST's Little Red Dots show blackbody-like continua, broad Balmer lines, and lack of variability. Similar features appear in Type IIIn supernovae and eta Carinae's 19th-century Great Eruption.

Are Little Red Dots overmassive black holes or something else? Their true nature remains unclear.

What this paper is selling

The authors propose that dense winds from a central engine create a pseudo-photosphere, mimicking a virial broad-line region. This explains the entire LRD life cycle, from enshrouded birth to black hole seeds.

Key results to remember

- LRDs may be intermediate-mass (10^3 - $6 M_{\text{sun}}$), super-Eddington engines, not overmassive black holes.
- Dense winds produce pseudo-photospheres, explaining blackbody continua and line profiles.
- Dust production in envelopes may mark the beginnings of classical AGN.

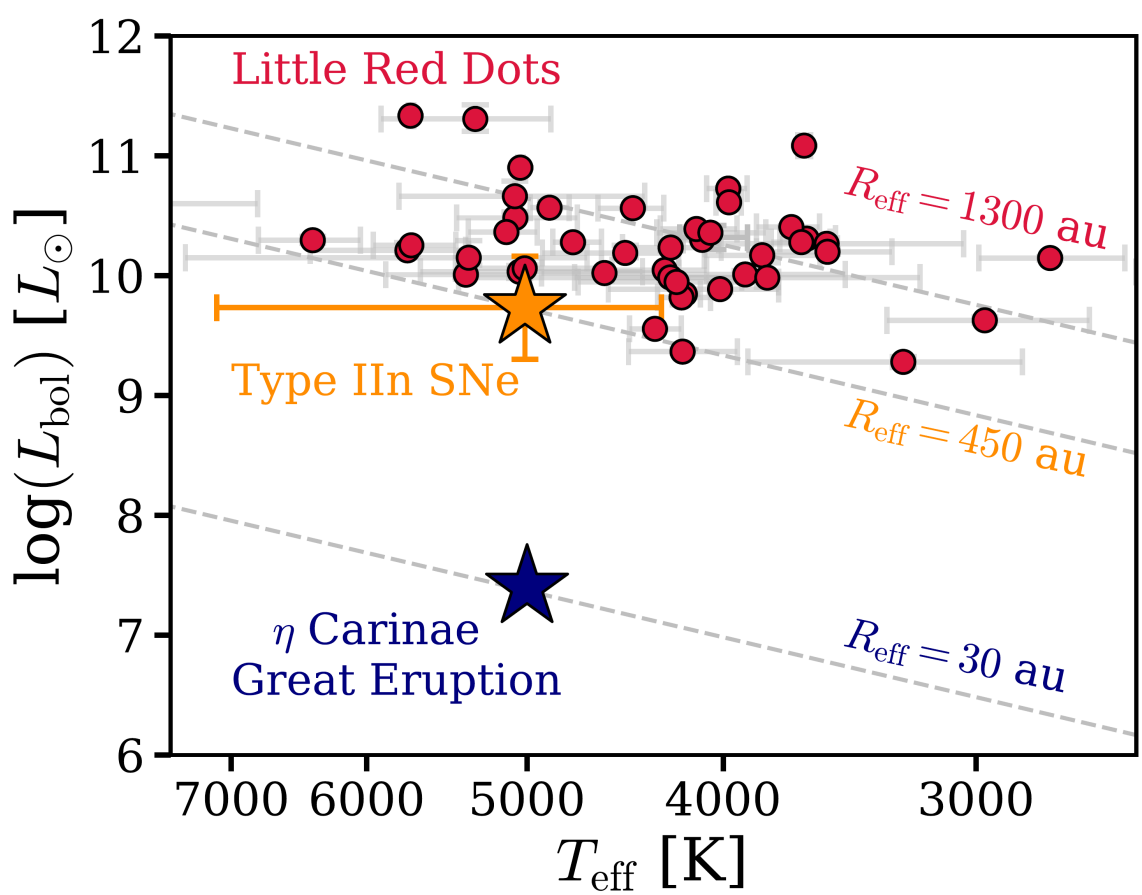


Figure 4 | LRDs, Type IIIn SNe, and eta Carinae's Great Eruption cluster at similar effective temperatures despite spanning 4 dex in luminosity.

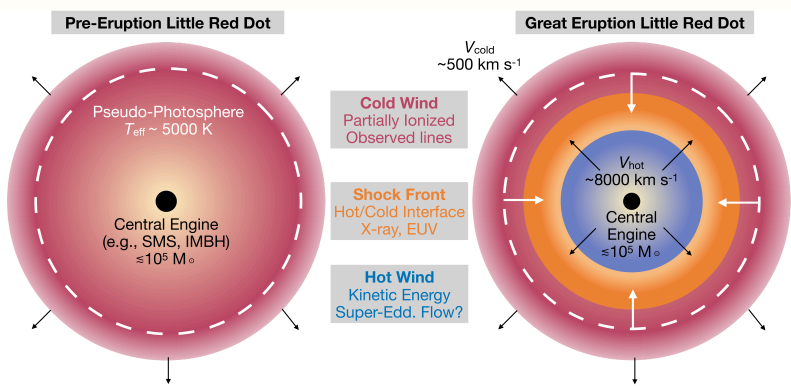


Figure 8 | Schematic shows how a cold wind creates a pseudo-photosphere, while hot wind shocks power the LRD's radiation.